

Healthcare Performance Management System - User Guide

System Overview

The Healthcare Performance Management System is a web-based platform designed to track, monitor, and analyze healthcare performance across 23 Woredas (districts) using a comprehensive 110-point scoring system. The system enables real-time data entry, automatic performance calculations, and dynamic reporting with interactive visualizations.

User Roles & Access

Department Heads (23 Users)

Each department head has exclusive access to their specific indicator data entry interface:

- **EPI Department:** Manages immunization program performance
- **TB & Leprosy Department:** Tracks tuberculosis and leprosy indicators
- **Child Health Department:** Monitors child health metrics
- **PHEM Department:** Manages public health emergency management
- **CBHI Department:** Handles community-based health insurance
- **Finance Department:** Oversees financial performance indicators



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- **Plan Department:** Manages planning and strategy metrics
- **WT Department:** Tracks workforce and training indicators
- **Medical Service Department:** Manages medical service delivery
- **RMH Department:** Responsible for referral and maternal health
- **Pharmacy & Logistic Department:** Oversees pharmaceutical services
- **Ultrasound Department:** Manages diagnostic imaging services
- **APTS Department:** Handles ambulatory patient treatment services
- **Community Pharmacy Department:** Manages community pharmacy operations
- **DM Test Department:** Oversees diabetes mellitus testing programs
- **Full EMR Department:** Manages electronic medical records implementation
- **EPI Modernization Department:** Tracks EPI system modernization
- **Zero Dose Department:** Monitors zero-dose children identification
- **Multi-Sectoral Department:** Coordinates multi-sector health initiatives
- **Cash Program Department:** Manages cash-based health programs
- **Hygiene Department:** Oversees hygiene and sanitation indicators



- **HIV/STI Department:** Manages HIV and STI program performance

Administrators (2 Users)

- **Admin:** Views all performance data, generates reports, analyzes trends
- **Super Admin:** Full system control including user management and database administration

System Architecture & Workflow

Dynamic Calculation Engine

The system automatically calculates performance scores in real-time using a sophisticated algorithm:

1. **Data Collection:** Each department enters scores for their specific indicators across all 23 Woredas
2. **Automatic Aggregation:** The system sums scores from all departments for each Woreda
3. **Percentage Calculation:** Converts total scores to percentages
($\text{score} \div 110 \times 100$)
4. **Real-time Ranking:** Automatically sorts Woredas by performance percentage
5. **Dynamic Updates:** Rankings and charts update instantly when new data is entered

Scoring Categories (110 Points Total)



Medical & Pharmacy (40 Points):

- Medical Service (15 pts): Assesses quality and accessibility of medical services
- RMH (10 pts): Evaluates referral and maternal health systems
- Pharmacy & Logistic (5 pts): Measures pharmaceutical supply chain efficiency
- Ultrasound (2.5 pts): Tracks diagnostic imaging availability
- APTS (2.5 pts): Monitors ambulatory treatment services
- Community Pharmacy (2.5 pts): Assesses community pharmacy coverage
- DM Test (2.5 pts): Evaluates diabetes screening programs

Prevention & Disease (20 Points):

- EPI (5 pts): Immunization program coverage and effectiveness
- Child Health (5 pts): Child health services and nutrition programs
- TB & Leprosy (5 pts): Disease detection and treatment programs
- PHEM (5 pts): Emergency response and disease surveillance

Admin & Finance (25 Points):

- CBHI (10 pts): Community-based health insurance enrollment and management
- Finance (5 pts): Financial management and resource allocation
- Plan (5 pts): Strategic planning and program implementation



- WT (5 pts): Workforce training and capacity building

Innovation & Quality (25 Points):

- Full EMR (5 pts): Electronic medical records system implementation
 - EPI Modernization (2.5 pts): Immunization system upgrades and innovations
 - Zero Dose (2.5 pts): Identification and outreach to unimmunized children
 - Multi-Sectoral (2.5 pts): Cross-sector collaboration initiatives
 - Cash Program (2.5 pts): Cash-based health incentive programs
 - Hygiene & Sanitation (5 pts): Water, sanitation, and hygiene programs
 - HIV/STI (5 pts): HIV and STI prevention and treatment services
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Data Entry Process

Department Head Workflow

Step 1: System Access

- Navigate to the web application URL
- Enter assigned username and password
- System automatically redirects to department-specific interface

Step 2: Data Entry Interface



- View department-specific data entry form
- See list of all 23 Woredas with input fields
- Each Woreda has a slider or number input field
- Maximum values are pre-set based on indicator importance
- Current values are displayed if data exists

Step 3: Score Input

- Use slider controls to enter performance scores
- Input values between 0 and maximum allowed for each indicator
- System validates input ranges in real-time
- Visual feedback shows score impact on overall performance

Step 4: Data Validation

- System automatically checks for data consistency
- Validates score ranges and formats
- Highlights any potential data entry errors
- Provides confirmation before saving

Step 5: Save & Update

- Click "Save [Department] Data" button
- System processes and stores data securely
- Real-time updates to rankings and charts
- Confirmation message displays successful save

Step 6: Review Results

- View updated performance rankings immediately
 - See impact of data entry on Woreda positions
 - Access detailed performance reports
 - Export data if needed
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Reporting & Analytics

Admin Dashboard Features

Key Performance Indicators (KPIs)

- **Average Score:** Mean performance across all Woredas
- **Highest Score:** Top-performing Woreda achievement
- **Total Woredas:** Number of Woredas with complete data
- **Performance Trends:** Historical performance patterns

Interactive Visualizations

- **Bar Charts:** Horizontal bar chart showing Woreda rankings with color-coded performance levels
- **Performance Trends:** Line graphs showing performance over time
- **Departmental Contributions:** Stacked charts showing each department's impact
- **Geographic Distribution:** Maps showing regional performance patterns



Detailed Analytics

- **Woreda Comparisons:** Side-by-side performance comparisons
- **Department Analysis:** Individual department performance metrics
- **Trend Analysis:** Performance changes over time periods
- **Benchmarking:** Performance against targets and standards

Report Generation

- **Real-time Reports:** Automatically generated when data updates
- **Custom Filters:** Filter by date range, Woreda, or department
- **Export Options:** Download reports in various formats
- **Scheduled Reports:** Automated report generation and distribution

User Interface Design

Navigation Structure

- **Sidebar Navigation:** Easy access to different system sections
- **Role-based Menus:** Customized navigation based on user role
- **Quick Actions:** One-click access to common tasks
- **Breadcrumb Trail:** Clear navigation path indication

Data Entry Interface



- **Intuitive Forms:** User-friendly input controls
- **Visual Feedback:** Real-time validation and guidance
- **Progress Indicators:** Show completion status
- **Help Text:** Contextual assistance for each field
- **Responsive Design:** Works seamlessly on all devices

Dashboard Layout

- **Card-based Design:** Organized information display
- **Color Coding:** Visual performance indicators (green/yellow/red)
- **Interactive Elements:** Clickable charts and data points
- **Responsive Grid:** Adapts to different screen sizes
- **Accessibility Features:** Compliance with accessibility standards

Mobile Experience

- **Touch-friendly Controls:** Optimized for mobile interaction
- **Responsive Layout:** Adapts to phone and tablet screens
- **Quick Access:** Essential functions readily available
- **Offline Capability:** Basic functionality without internet

System Operations

Real-time Processing

- **Instant Calculations:** Scores update immediately upon data entry



- **Live Rankings:** Woreda positions change dynamically
- **Automatic Refresh:** Charts and graphs update without manual refresh
- **Concurrent Access:** Multiple users can enter data simultaneously

Data Management

- **Secure Storage:** All data encrypted and backed up
- **Version Control:** Track changes and maintain data history
- **Data Integrity:** Automatic validation and error correction
- **Backup Systems:** Regular data backups and recovery options

Performance Optimization

- **Fast Loading:** Optimized for quick page loads
- **Efficient Queries:** Database optimization for rapid data retrieval
- **Caching System:** Frequently accessed data stored locally
- **Scalability:** Handles increasing data volumes efficiently

Graph Types & Visualizations

Performance Rankings

- **Horizontal Bar Chart:** Shows Woreda rankings with performance scores
- **Color Coding:** Green (80-100%), Yellow (50-79%), Red (0-49%)



- **Interactive Tooltips:** Hover for detailed information
- **Sorting Options:** Sort by score, Woreda name, or department

Departmental Analysis

- **Stacked Bar Charts:** Show each department's contribution to total scores
- **Pie Charts:** Display departmental weight distribution
- **Trend Lines:** Show performance changes over time
- **Comparison Charts:** Side-by-side department performance

Geographic Visualization

- **Regional Maps:** Performance by geographic area
- **Heat Maps:** Intensity of performance across regions
- **Cluster Analysis:** Grouping of similar-performing Woredas
- **Distance Analysis:** Performance variation by location

Time Series Analysis

- **Line Graphs:** Performance trends over time periods
- **Moving Averages:** Smoothed performance trends
- **Seasonal Patterns:** Cyclical performance variations
- **Forecasting:** Predictive performance modeling



Personalization

- **Custom Dashboards:** User-specific view configurations
- **Saved Preferences:** Remember user settings and layouts
- **Quick Links:** Frequently accessed functions
- **Recent Activity:** Track recent data entries and changes

Collaboration Tools

- **Comments System:** Add notes to data entries
- **Approval Workflows:** Multi-level data validation
- **Notification System:** Alerts for important updates
- **Activity Logs:** Track all system activities

Accessibility

- **Screen Reader Support:** Compatible with assistive technologies
- **Keyboard Navigation:** Full keyboard accessibility
- **High Contrast Mode:** Enhanced visibility options
- **Language Support:** Multi-language capability

System Benefits

For Department Heads

- **Easy Data Entry:** Intuitive interface for score input
- **Immediate Feedback:** Real-time performance impact



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- **Standardized Process:** Consistent data entry across departments
- **Mobile Access:** Enter data from any location

For Administrators

- **Comprehensive Overview:** Complete system visibility
- **Data-driven Decisions:** Evidence-based management
- **Performance Monitoring:** Real-time tracking of indicators
- **Report Generation:** Automated reporting capabilities

For Healthcare System

- **Improved Quality:** Better healthcare service delivery
- **Accountability:** Clear performance measurement
- **Resource Optimization:** Efficient resource allocation
- **Strategic Planning:** Data-informed decision making

📞 Support & Maintenance

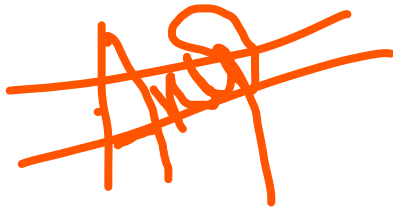
Technical Support

- **24/7 Availability:** Continuous system access
- **User Training:** Comprehensive training materials
- **Help Documentation:** Detailed user guides and tutorials
- **Contact Support:** Direct assistance via Telegram (@AI_Technology)

System Updates

- **Regular Updates:** Continuous system improvements
- **Feature Enhancements:** New functionality additions
- **Security Updates:** Ongoing security improvements
- **Performance Optimization:** System speed and reliability enhancements

 This comprehensive Healthcare Performance Management System provides a powerful, user-friendly platform for monitoring and improving healthcare delivery across all Woredas, enabling data-driven decision making and continuous quality improvement.

A handwritten signature in orange ink, consisting of stylized, overlapping letters, is centered within a large, irregular green border.